FarmRoomba
Cotton-Harvesting Robot

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Project Objectives and Goals

Successfully navigate a simulated cotton arena while harvesting ripe cotton bolls with minimal foreign material.

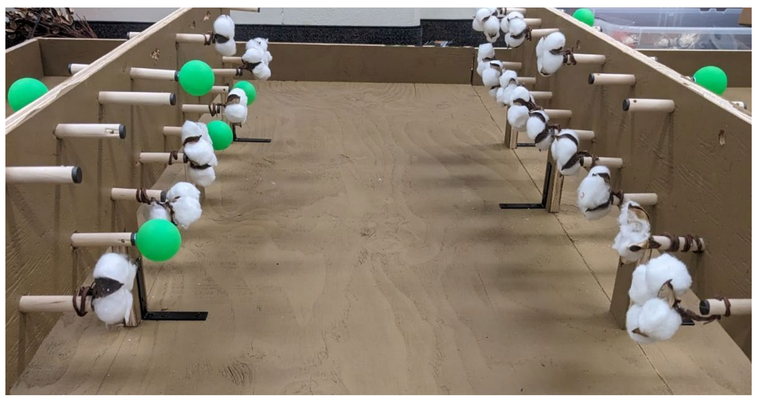


Figure 1. Mock competition layout

Background

Industry money is lost because cotton crop degrades due to adverse weather and delayed harvesting. An autonomous robot can selectively harvest the cotton as it becomes ripe.

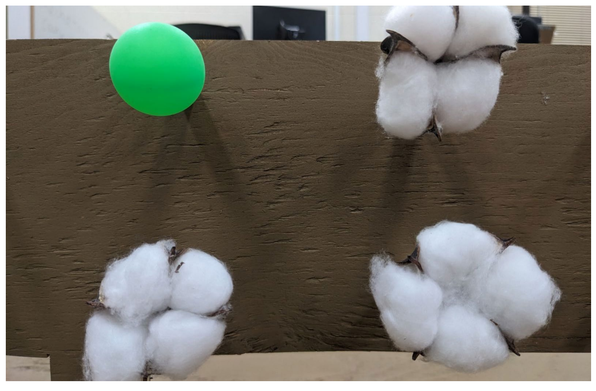


Figure 2. Sample layout of ripe and unripe cotton bolls on the arena wall

Harvesting Apparatus

The robot pulls cotton off the boll using negative pressure from an onboard vacuum. The maximum force is applied to each boll through a rotating orifice plate, which directs the opening at each ball in turn.

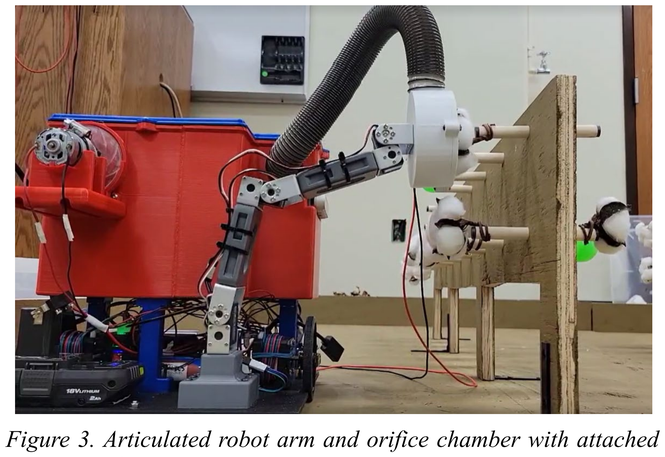


Figure 3. Articulated robot arm and orifice chamber with attached vacuum and storage.

Sensor Overview

The robot uses a combination of depth images and color maps from an Intel RealSense camera to get its position relative to the wall. This also allows the robot to simultaneously identify ripe and unripe cotton bolls.

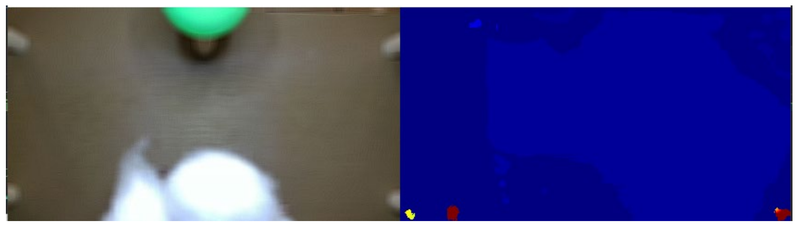


Figure 4. Sample color and depth images captured with the RealSense camera



Figure 5. Green filter applied

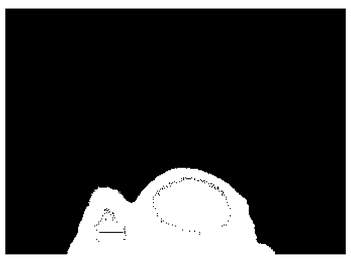


Figure 6. White filter applied

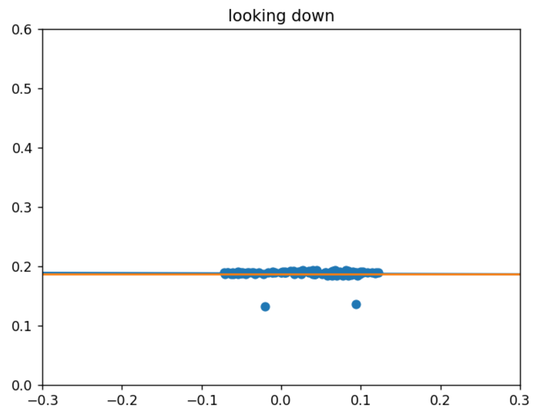


Figure 7. Orientation vector generated using a depth image from the RealSense camera

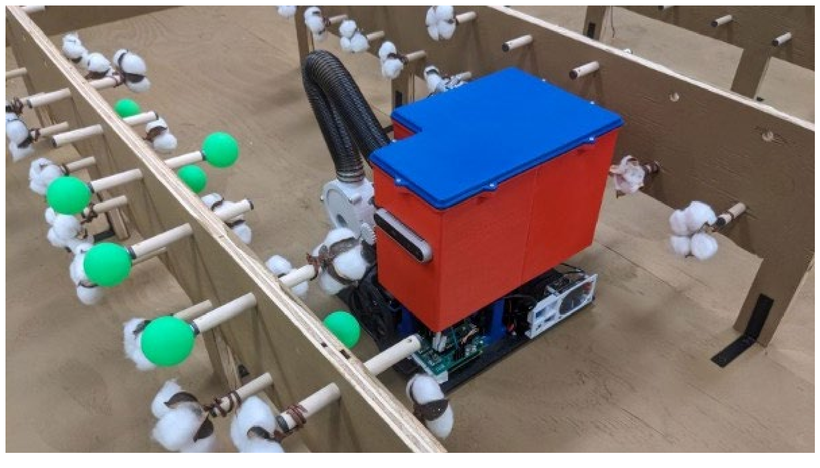


Figure 8. Depiction of the placement of the RealSense camera in the robot

Software Architecture

The entire robot operates within a state machine that enables each function to be called at the appropriate time. This enables mapping, harvesting, and navigation to be carried out in proper order for each boll.

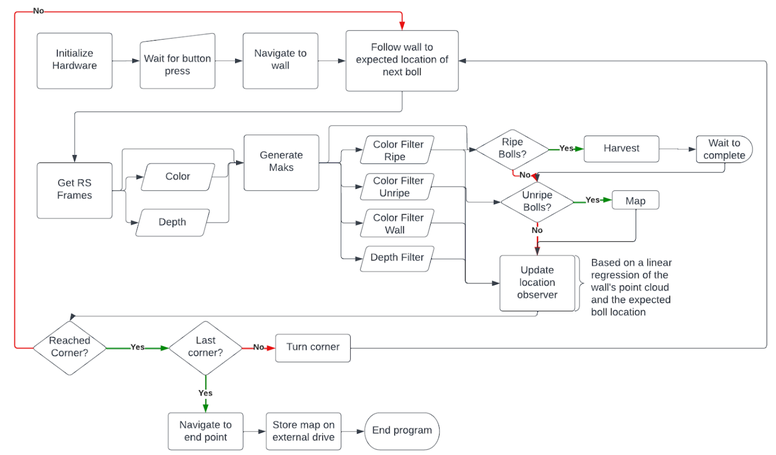


Figure 9. State machine outlining the control sequence of the robot

Future Improvements

Given this was the first year BYU entered a team, there are a few items we plan to improve on in the future.

- Reduce the size and footprint of the stepper drivers to incorporate them directly into the PCB
- Reduce the criticality of single components to improve the overall reliability of the robot
- Consolidate the power systems of the robot to operate on a
- Begin integrating subsystems at an earlier stage to facilitate fewer compounding problems during final assembly

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Our team would like to recognize the generous grant we received from the BYU Weidman Center that has made it possible for us to participate in this year's competition.